

Aarush Khare

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Education

University of California, Berkeley

June 2025 – Present

B.A. in Computer Science, CDSS | GPA: 4.0/4.0

Berkeley, CA

Honors: Regents' and Chancellor's Scholar, Dean's List (Fall 2025)

Coursework:

- **Systems & Theory:** CS 61C (Machine Structures), CS 170 (Algorithms), CS 47B (Data Structures), CS 70 (Discrete Math), CS 61A (Computer Programming)
- **AI & Machine Learning:** CS 189 (Machine Learning), CS 188 (Intro to AI), CS 198 (Computer Vision)

Experience

Undergraduate Researcher: Berkeley Operations and Behavioral Analytics Lab

January 2026 – Present

- Building software infrastructure and optimization pipelines for cost-constrained 3D design generation.
- Integrating learned cost-estimation generative AI models with constrained search and iterative design refinement.
- Professor: Park Sinchaisri

Research Assistant: Abbasi Lab (Computational Neuroscience at UCSF)

January 2026 – Present

- Developing advanced machine learning algorithms to map and analyze neuron functionality in the brain.
- Correlating spatial neuron data with spatially resolved gene expression profiles to uncover structural patterns.
- Professor: Reza Abbasi-Asl

Computer Science Instructor: AlphaStar Academy

August 2024 – May 2025

Taught algorithmic concepts, prepared lecture notes, and guided practice for students using Python, C++, and Java.

Math Workshop Director: Avenidas Senior Center

October 2023 – June 2025

Created and delivered math-based workshops for seniors to promote mental sharpness and slow cognitive decline.

Achievements

William Lowell Putnam Math Competition: Top 500

February 2026

Ranked 312th nationwide in the premier undergraduate math competition.

USA Computing Olympiad (USACO): Platinum Division

December 2023

Ranked 69th among pre-college participants worldwide in January 2024.

USA Math Olympiad (USAMO): 2x USAJMO Winner, 1x USAMO HM

2022, 2023, 2024

Top 25 students 10th grade and under in the United States, Top 125 high school students.

Future Business Leaders of America (FBLA): 4th Place National Rank

April 2023, 2024

Ranked 4th in Coding & Programming in the United States; 1st place in Data Analysis, State Champion.

PROMYS (Math Program) at Boston University: Student

June 2024 – August 2024

Program in Mathematics for Young Scientists; One of 80 students selected worldwide to attend an intensive, residential math program.

Harvard-MIT Math Tournament (HMMT): 2nd Place Overall Team

February 2024

Stanford Math Tournament (SMT): 2nd Place Overall Team

April 2023

Skills

C++, Java, Python, ReactJS, NextJS, Typescript, C, RISC-V, SQL, $\LaTeX$, Asymptote